

BS1a

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Outline for Week 7

- Goodness of fit for GLM
- GLM diagnostics
- Contingency tables

Goodness of fit (GOF) for GLM

We have seen how to select the best model out of a list of models but does this “best” model fit the data adequately?

The best model fit is achieved by the saturated model. Since there is one parameter for each response, i.e. no constraints are imposed on the parameters, there is not misfit.

When a model simpler than the saturated model is fitted, there will be a decrease in the value of the maximised log-likelihood. But is this decrease significant?

The **(residual) deviance** $D(y)$ is the **likelihood ratio test statistic** comparing the saturated model with n parameters to the model of interest with p parameters where $p < n$. The distribution of $D(y)$ is **approximately** χ^2_{n-p} .

For Binomial and Poisson models (amongst others) it can be shown that D can be directly estimated by the data

$$\text{Poisson : } D = 2 \left[\sum_{i=1}^n y_i \log(y_i/\hat{y}_i) - \sum_{i=1}^n (y_i - \hat{y}_i) \right]$$

$$\text{Binomial : } D = 2 \left[\sum_{i=1}^n y_i \log(y_i/\hat{y}_i) + \sum_{i=1}^n (n_i - y_i) \log \left(\frac{n_i - y_i}{n_i - \hat{y}_i} \right) \right]$$

Proof

The approximation of the distribution of D works well when the relevant binomial or poisson distributions are reasonably approximated by normal distributions i.e. large number of trials in the binomial case and fitted counts greater than 5 for the poisson case.

In practice, these approximations can be quite poor for sparse data.

Davison: *The problem is most acute for the deviance of binary data for which the large-sample approximation is useless.*

In the Bernoulli case, $n_i = 1$ and $y_i = 0$ or 1 and the deviance reduces to:

$$-2 \sum_{i=1}^n \left\{ \hat{p}_i \log \left(\frac{\hat{p}_i}{1 - \hat{p}_i} \right) + \log(1 - \hat{p}_i) \right\}$$

therefore, it only depends on the fitted values and not on the data so it cannot be used as a GOF test.

Going back to the budworms example.

We saw that the additive model with ldose and sex as explanatory variables did not have any redundant parameters in it, but does it fit the data?

Compare to the saturated model (from **R** output) $D(y) = 6.757$ while $n = 12$ and $p = 3$, therefore the p-value is 0.67.

High p-values indicate no evidence of misfit of the model of interest.

On the other hand, consider the model with ldose as the only explanatory variable and compare to the saturated model.

In this case there is mild evidence of misfit (p-value=0.075) which is not surprising since we know that sex is a significant explanatory variable.

When fitted values are very small the distribution of the GOF statistic is not approximately χ^2 . Similarly, residuals have limited meaning in that case. For ungrouped binary data and often when explanatory variables are continuous each cell count is equal to 1. Then y_i can equal only 0 and 1 and a residual can assume only two values and is usually uninformative. Plots of residuals also can have limited use consisting merely of two parallel lines of dots. The deviance itself is then completely uninformative about model fit. When data can be grouped into sets of observations having common predictor values it is better to compute residuals for the grouped data than for individual values.

GLM diagnostics Further insight about the fit of the model is obtained by comparing observed and fitted values individually:

- **Deviance Residuals**, in **R**: `residuals()`,

$$r_i = \text{sign}(y_i - \hat{\mu}_i) \sqrt{d_i}$$

where $\text{sign}(y_i - \hat{\mu}_i)$ is +1 or -1 depending on the sign of $y_i - \hat{\mu}_i$ and shows us whether y_i is in conflict by being too large or too small compared to the fitted mean and d_i is the contribution of the point to the deviance of the model:

$$d_i = 2(\ell_i(\hat{\theta}_i^{(s)}; y_i) - \ell_i(\hat{\beta}; y_i)), \quad \sum_i d_i = D(y)$$

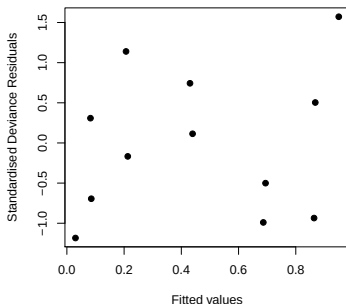
- **Standardised Deviance Residuals**, in **R**: `rstandard()`,

$$r'_i = \frac{r_i}{\sqrt{1 - h_i}}$$

where h_i are the leverages. Under the assumption that the fitted model is correct the r' have roughly unit variance and distribution close to normal.

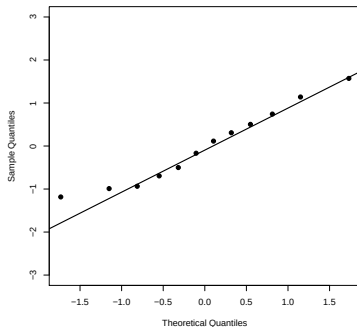
Going back to the budworm moths example.

The latter can be plotted against the fitted values:



Absolute values of the standardised deviance residuals greater than 2 or 3, any pattern eg. curvature and increase in the spread of the residuals for large fitted values (funneling) are indications of a misfit.

And a QQ-plot:



Very few data points but probably looks acceptable.

Points of influence

As in the case of the normal linear model, the **leverages** are given by the diagonal of the projection matrix H .

In GLM, H is no longer a function of X alone because of the weights used in the IWLS algorithm.

In this case $H = W^{1/2}X(X^TWX)^{-1}X^TW^{1/2}$ and therefore the leverages also depend on y .

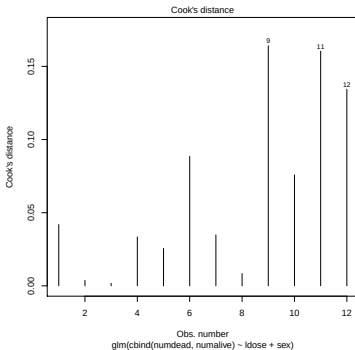
Remember that $\text{tr}(H) = p$ so the average leverage cannot be reduced below p/n .

In this case there are no points with high leverage.

To measure the **influence** of a point we can examine the Cook's distances:

$$C_k = \frac{(\hat{\beta} - \hat{\beta}_{-k})^T (X^T W X) (\hat{\beta} - \hat{\beta}_{-k})}{p\phi}$$

in **R**: `cooks.distance(glm.add)`



Contingency tables

Consider the following table:

	recovered		Total
	Y	N	
treated	n_{11}	n_{12}	n_{1+}
untreated	n_{21}	n_{22}	n_{2+}
Total	n_{+1}	n_{+2}	n

The cell counts $n_{11}, n_{12}, n_{21}, n_{22}$ are realisations of random variables but what are the distributions of these variables?

It depends on the way the data are collected.

- **Fixed marginal totals**

- Prospective sampling i.e. the predictors are fixed and then the outcome is observed (fixed row totals)

Suppose that it was decided that the sample would consist of n_{1+} treated and n_{2+} untreated individuals. Then

$n_{ij} \sim \text{Binomial}(n_{i+}, \pi_{ij})$. Constraints: $\sum_j n_{ij} = n_{i+}$ and $\sum_j \pi_{ij} = 1, \forall i$.

- Retrospective sampling i.e. the outcome is fixed and then the predictors are observed (fixed column totals)

Suppose that it was decided that the sample would consist of n_{+1} recovered and n_{+2} non-recovered individuals. Then

$n_{ij} \sim \text{Binomial}(n_{+j}, \pi_{ij})$. Constraints: $\sum_i n_{ij} = n_{+j}$ and $\sum_i \pi_{ij} = 1, \forall j$.

Example from Agresti: 1127 Americans are classified according to their gender and their belief in afterlife:

	Belief in Afterlife		Total
	Yes	No/Undecided	
Females	$n_{11} = 509$	$n_{12} = 116$	$n_{1+} = 625$
Males	$n_{21} = 398$	$n_{22} = 104$	$n_{2+} = 502$
Total	$n_{+1} = 907$	$n_{+2} = 220$	$n = 1127$

1 Fixed row totals

The question is: **Does sex predict belief?**

```
data1 <- data.frame(Y=c(509,398),N=c(116,104),Sex=c("F","M"))
glm.fit1 <- glm(cbind(data1$Y,data1$N) ~ Sex, family = binomial, data=data1)
summary(glm.fit1)
```

```
##Call:
##glm(formula = cbind(data1$Y, data1$N) ~ Sex, family = binomial,
##     data = data1)
```

```
##Coefficients:
```

##	Estimate	Std. Error	z value	Pr(> z)
##(Intercept)	1.4789	0.1029	14.374	<2e-16 ***
##SexM	-0.1368	0.1507	-0.908	0.364
##---				

(Odds for believing in afterlife for males)/(Odds for believing in afterlife for females) = $\exp(\hat{\beta}_2) = \exp(-0.1368) = 0.8721$ (but according to Wald test this is not significantly different from 1.)

2 Fixed column totals

The question is: **Does belief predict sex?**

```
data2 <- data.frame(M=c(104,398),F=c(116,509),Belief=c("N","Y"))
glm.fit2 <- glm(cbind(data2$M,data2$F) ~ Belief, family = binomial, data=data2)
summary(glm.fit2)
```

```
##Call:
##glm(formula = cbind(data2$M, data2$F) ~ Belief, family = binomial,
##     data = data2)
```

```
##Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
##(Intercept)  -0.1092     0.1350  -0.809   0.419
##BeliefY       -0.1368     0.1507  -0.908   0.364
```

(Odds for being a male for believers)/(Odds for being a male for non-believers) = $\exp(\hat{\beta}_2) = \exp(-0.1368) = 0.8721$.

The same as before!

For both cases, the odds of being a male believer divided by the odds of being a female believer are:

$$\frac{\frac{\pi_{21}}{\pi_{22}}}{\frac{\pi_{11}}{\pi_{12}}} = \frac{\pi_{21}\pi_{12}}{\pi_{11}\pi_{22}}$$

1

$$\frac{\pi_{21}\pi_{12}}{\pi_{11}\pi_{22}} = \frac{\frac{n_{21}}{n_{2+}} \frac{n_{12}}{n_{1+}}}{\frac{n_{11}}{n_{1+}} \frac{n_{22}}{n_{2+}}} = \frac{n_{21}n_{12}}{n_{11}n_{22}}$$

2

$$\frac{\pi_{21}\pi_{12}}{\pi_{11}\pi_{22}} = \frac{\frac{n_{21}}{n_{+1}} \frac{n_{12}}{n_{+2}}}{\frac{n_{11}}{n_{+1}} \frac{n_{22}}{n_{+2}}} = \frac{n_{21}n_{12}}{n_{11}n_{22}}$$

- **Fixed overall total**

Suppose that it was decided that the sample would consist of n individuals. Then $n_{ij} \sim \text{Multinomial}(n, \pi_{ij}), \forall i, j$. Constraints: $\sum_{i,j} n_{ij} = n$ and $\sum_{i,j} \pi_{ij} = 1$.

- **No constraints on the totals**

Suppose that it was decided that all cases that would appear in a given period would be part of the sample. Then $n_{ij} \sim \text{Poisson}(\lambda_{ij}), \forall i, j$ and there are no constraints placed on the totals.

Log-linear models

All of the four cases presented before can be modelled using the general class of **log-linear models**. The name refers to the canonical link of a Poisson GLM which is the log-link.

When constraints are placed on the totals (either row or column) then the parameters associated with the fixed margins need to be incorporated in the model.

For contingency tables what is usually of interest are the associations between variables which in log-linear models are expressed using the interaction terms.

Going back to the Americans

The questions “ Does sex predict belief” and “Does belief predict sex” can be answered by testing the null hypothesis that the interaction term is equal to 0 in the saturated model

$$\log(E(Y_i)) = \beta_1 + \beta_2 \text{sex}_i + \beta_3 \text{belief}_i + \beta_4 \text{sex}_i \text{belief}_i$$

where $\text{sex}_i=1$ for males and $\text{belief}_i=1$ for believers.

The expected counts in each cell are:

		Belief in Afterlife	
Females	$\hat{n}_{11} = \exp(\hat{\beta}_1 + \hat{\beta}_3)$	$\hat{n}_{12} = \exp(\hat{\beta}_1)$	
Males	$\hat{n}_{21} = \exp(\hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_3 + \hat{\beta}_4)$	$\hat{n}_{22} = \exp(\hat{\beta}_1 + \hat{\beta}_2)$	

Therefore, the estimated odds ratio is:

$$\frac{\hat{n}_{21}\hat{n}_{12}}{\hat{n}_{11}\hat{n}_{22}} = \frac{\exp(\hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_3 + \hat{\beta}_4) \exp(\hat{\beta}_1)}{\exp(\hat{\beta}_1 + \hat{\beta}_3) \exp(\hat{\beta}_1 + \hat{\beta}_2)} = \exp(\hat{\beta}_4)$$

```
data3 <- data.frame(Y=c(509,116,398,104),
Belief=c("Y","N","Y","N"),Sex=c("F","F","M","M"))
glm.fit3 <- glm(Y~Sex*Belief, family = poisson, data=data3)
summary(glm.fit3)
```

```
##Call:
```

```
##glm(formula = Y ~ Sex * Belief,
##family = poisson, data = data3)
```

```
##Coefficients:
```

##	Estimate	Std. Error	z value	Pr(> z)	
##(Intercept)	4.75359	0.09285	51.198	<2e-16	***
##SexM	-0.10920	0.13504	-0.809	0.419	
##BeliefY	1.47886	0.10288	14.374	<2e-16	***
##SexM:BeliefY	-0.13680	0.15071	-0.908	0.364	
##---					

where we obtain the same result as before.